Class Diagram: ระบบประกอบและจัดจำหน่ายคีย์บอร์ดคัสตอมออนไลน์ KeyCraft

User
- # id: ObjectId
- # name: String
- # email: String
- # password: String
- # phone: String
- # avatar: String
- # role: String
- # lastLogin: Date

+ login(email, pwd): User
+ updateProfile(data): User
+ matchPassword(pwd): Boolean

Category
- - id: ObjectId
- - name: String
- - slug: String
- - icon: String
- - isActive: Boolean

+ getProducts(): Product[]

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Product
- - id: ObjectId
- - name: String
- - category: String
- - brand: String
- - price: Number
- - stock: Number
- - inStock: Boolean
- - image: String
- - partType: String
- - layout: String
- - switchType: String
- - mountType: String

+ getProducts(): Product[]
+ isAvailable(): Boolean
+ deductStock(qty): void
+ updateDetails(data): void

Customer

+ createOrder(cart): Order
+ uploadPaymentProof(id, slip): Boolean
+ cancelOrder(id): Boolean
+ register(userData): User

Employee
- - employeeId: String
- - accountType: String

+ updateOrderStatus(id, status): Order
+ manageProduct(productData): Product
+ viewReports(): ReportData
+ viewSlip(orderId): String

Address
+ + street: String
+ + subdistrict: String
+ + district: String
+ + province: String
+ + postalCode: String
+ + phone: String

+ getFullAddress(): String

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OrderItem
- - name: String
- - price: Number
- - quantity: Number
- - specs: String
- - image: String
- - switchSound: String

+ getItemTotal(): Number

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Part
- - type: String (case|switch|plate|keycap)
- - name: String
- - color: String
- - secondaryColor: String
- - price: Number
- - image: String

+ getPartInfo(): String

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Order
- - orderId: String
- - shippingAddress: Address
- - orderType: String (prebuilt | custom)
- - status: OrderStatus
- - customerInfo: CustomerInfo
- - subtotal: Number
- - discount: Number
- - shipping: Number
- - total: Number
- - trackingNumber: String
- - paymentProof: String
- - createdAt: Date

+ calculateTotal(): Number
+ updateStatus(newStatus): void
+ lockOrder(): void
+ generateReceipt(): Receipt

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Receipt
- - method: String = "PromptPay"
- - promptPayAccount: String
- - qrExpiresAt: Date
- - verifiedAmount: Number
- - transactionId: String
- - isVerified: Boolean

+ generatePromptPayQr(amount): QrData
+ verifySlipWithSlipOK(slip): SlipResult

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Receipt
- - receiptNo: String
- - orderId: String
- - customerName: String
- - issueDate: Date
- - totalAmount: Number

+ printReceipt(): void
+ exportPdf(): Blob

Payment
- - method: String = "PromptPay"
- - promptPayAccount: String
- - qrExpiresAt: Date
- - verifiedAmount: Number
- - transactionId: String
- - isVerified: Boolean

+ generatePromptPayQr(amount): QrData
+ verifySlipWithSlipOK(slip): SlipResult